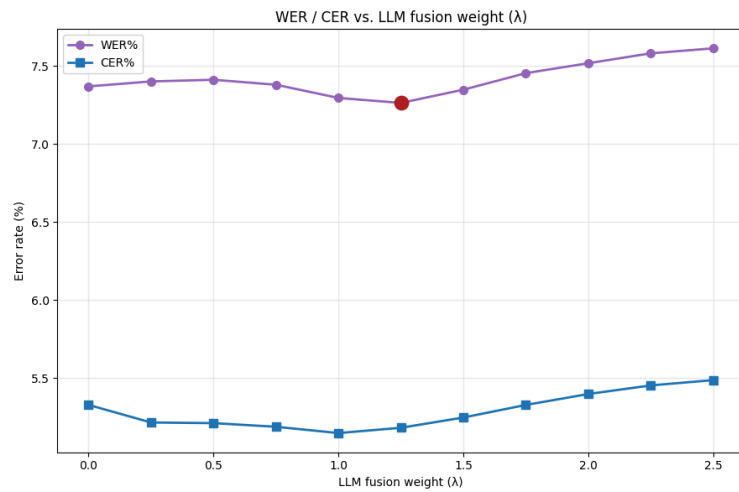
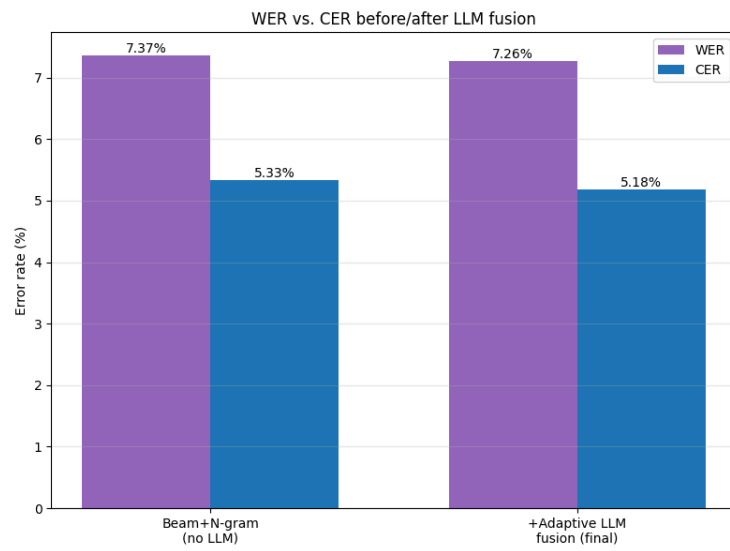
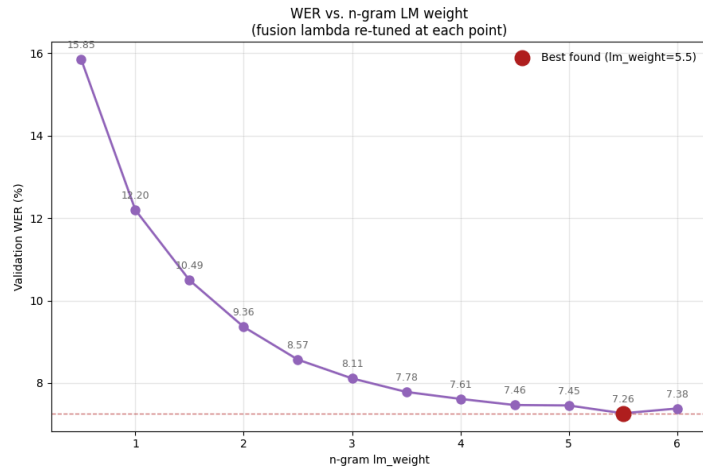


Qwen2.5-7B

===== SUMMARY =====

lm_weight	lambda	WER%	CER%	
0.5	2.50	15.85	10.13	
1.0	2.50	12.20	8.00	
1.5	2.50	10.49	7.04	
2.0	1.75	9.36	6.30	
2.5	1.75	8.57	5.86	
3.0	1.50	8.11	5.58	
3.5	1.25	7.78	5.42	
4.0	0.50	7.61	5.31	
4.5	0.50	7.46	5.23	
5.0	1.25	7.45	5.24	
5.5	1.25	7.26	5.18	<-- best
6.0	0.00	7.38	5.40	



Qwen3-4b :

===== SUMMARY =====

lm_weight	lambda	WER%	CER%	
0.5	2.5	16.20	10.22	
1.0	2.0	12.58	8.23	
1.5	1.25	10.85	7.18	
2.0	1.5	9.62	6.48	
2.5	0.75	8.72	5.94	
3.0	1.25	8.30	5.71	
3.5	0.0	7.87	5.46	
4.0	0.0	7.71	5.43	
4.5	0.0	7.55	5.38	
5.0	0.0	7.55	5.39	
5.5	0.0	7.37	5.33	<-- best
6.0	0.0	7.38	5.40	

Final validation numbers:

WER [Beam+NGram]: 7.37%

WER [Beam+NGram+LLM]: 7.37%

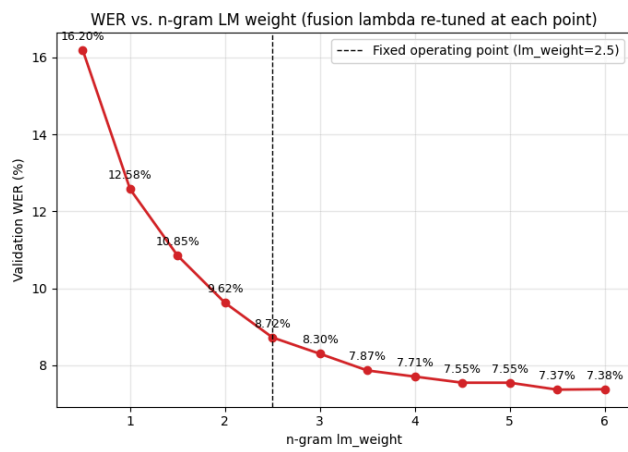
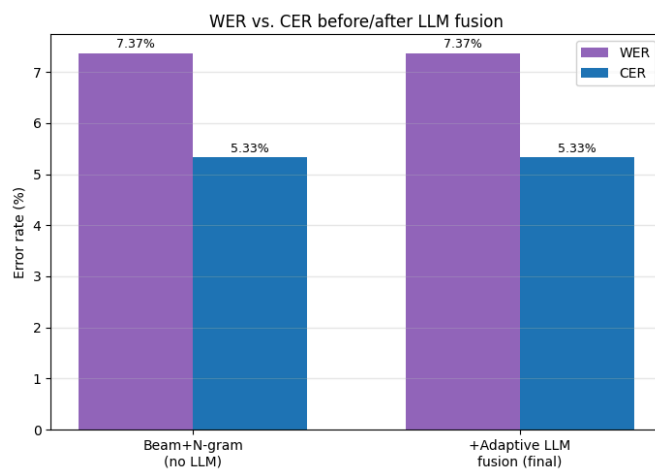
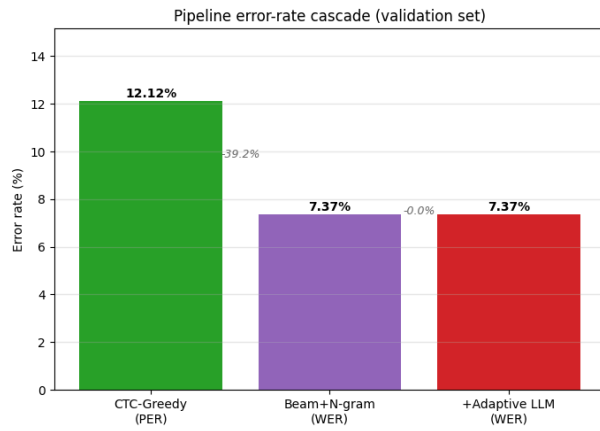
CER [Beam+NGram]: 5.33%

CER [Beam+NGram+LLM]: 5.33%

PER [CTC-Greedy]: 12.12%

Overall metrics:

Metric	Beam+NGram (%)	Beam+NGram+LLM (%)
WER	7.368978	7.368978
CER	5.328642	5.328642
PER	12.123116	12.123116



llama3.2-3b-instruct

===== SUMMARY =====

lm_weight	lambda	WER%	CER%	
0.5	2.5	15.67	9.97	
1.0	1.5	12.29	8.02	
1.5	1.0	10.64	7.07	
2.0	1.0	9.41	6.33	
2.5	1.25	8.63	5.86	
3.0	1.0	8.14	5.60	
3.5	1.0	7.80	5.42	
4.0	0.75	7.65	5.31	
4.5	0.75	7.50	5.25	
5.0	0.75	7.52	5.22	
5.5	0.5	7.33	5.16	<-- best
6.0	0.5	7.36	5.25	

Gemma3-4b

===== SUMMARY =====

lm_weight	lambda	WER%	CER%	
0.5	1.25	17.19	10.78	
1.0	1.25	13.41	8.72	
1.5	0.0	11.52	7.65	
2.0	0.75	10.19	6.85	
2.5	0.25	9.43	6.34	
3.0	0.0	8.91	6.15	
3.5	0.0	8.51	5.92	
4.0	0.0	8.33	5.87	<-- best
4.5	0.0	8.35	5.90	
5.0	0.0	8.40	6.02	
5.5	0.0	8.37	6.06	
6.0	0.0	8.37	6.10	

Mistral-7b

===== SUMMARY =====

lm_weight	lambda	WER%	CER%	
0.5	1.25	16.61	10.47	
1.0	1.25	13.16	8.57	
1.5	1.25	11.30	7.45	
2.0	0.75	10.04	6.78	
2.5	0.75	9.25	6.28	
3.0	1.25	8.80	6.08	
3.5	0.75	8.46	5.86	
4.0	0.5	8.27	5.75	<-- best
4.5	0.75	8.30	5.83	
5.0	0.5	8.35	5.91	
5.5	0.0	8.37	6.06	
6.0	0.25	8.33	5.98	

Final validation numbers:

WER [Beam+NGram]: 11.52%

WER [Beam+NGram+LLM]: 8.27%

CER [Beam+NGram]: 7.65%

CER [Beam+NGram+LLM]: 5.75%

PER [CTC-Greedy]: 12.12%

Overall metrics:

Metric	Beam+NGram (%)	Beam+NGram+LLM (%)
WER	11.519322	8.268925
CER	7.647470	5.752559
PER	12.123116	12.123116

